

□ Recall from last lecture:

Defn A probability space is a tuple $(\Omega, \mathcal{F}, \mathbb{P})$ where:

- Ω is a non-empty set ($\sim$ Sample space).
- $\mathcal{F}$ is a σ -algebra on Ω , i.e., $\mathcal{F}$ is a collection of subsets of Ω s.t.
 - $\emptyset, \Omega \in \mathcal{F}$
 - $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$
 - $\{A_n\}_{n=1}^{\infty}$ s.t. each $A_n \in \mathcal{F} \Rightarrow \bigcup A_n \in \mathcal{F}$.
- $\mathbb{P}: \mathcal{F} \rightarrow [0,1]$ is called a probability measure on $(\Omega, \mathcal{F})$ s.t.

- $\mathbb{P}(\Omega) = 1$

Countable additivity [If $\{A_n\}_{n=1}^{\infty}$ is a collection of pairwise disjoint sets each of which is in $\mathcal{F}$, then

$$\mathbb{P}\left(\bigcup A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

Rmk: We can ask for / talk about the probability of a set A only if $A \in \mathcal{F}$.

[2] For $\Omega \neq \emptyset$ set and $S \subseteq 2^\Omega$, $\sigma(S)$ is the sigma algebra generated by S , i.e., it is the smallest sigma algebra containing S .

[3] Defn. (Borel sigma algebra) Suppose (X, τ) is a topological space. Then the Borel σ -alg. of (X, τ) is defined to be σ -alg. generated by all open subsets of X .

$$\mathcal{B}(X) := \sigma(\tau).$$

Exercise Let (X, d) be a metric space which is separable (i.e., $\exists$ a countable dense subset of X). Then

$$\mathcal{B}(X) = \sigma(\{B(x, r)\}_{x \in X, r > 0}).$$

Soln. Since $\{B(x,r)\}_{x,r} \subseteq \mathcal{Z}$, we have

$$\sigma(\{B(x,r)\}_{x,r}) \subseteq \sigma(\mathcal{Z}) = \mathcal{B}.$$

To show the other way, it suffices to prove that every open set V :

$$V \in \sigma(\{B(x,r)\}_{x,r}).$$

For this, it suffices to show that V is a union of countably many balls. Say $\mathcal{Q} \subseteq X$ count. dense

Claim: $V = \bigcup_{i \in I} B(q_i, r_i)$, where $\{q_i\}_i = \mathcal{Q} \cap V$,
and $r_i > 0$ is the largest # s.t.
 $B(q_i, r_i) \subseteq V$.

Pf. of Claim

Suppose claim is false,

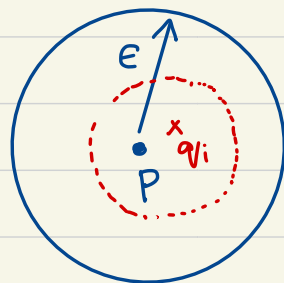
then $\exists p \in V \setminus \bigcup_i B(q_i, r_i)$.

If $r_i = \infty$, we think of

$$B(q_i, r_i) = \bigcup_{n=1}^{\infty} B(q_i, n)$$

Let $\epsilon > 0$ be s.t. $B(p, \epsilon) \subseteq V$. Since $\mathcal{Q}$ is dense, $\exists q_i \in B(p, \frac{\epsilon}{10})$, and hence

$$p \in B(q_i, \frac{\epsilon}{2}) \subseteq V, \text{ so } x_i > \frac{\epsilon}{2}.$$



Thus

$$p \in B(q_i, x_i). \quad (\Rightarrow \Leftarrow)$$

This proves the claim & completes the soln.

Rmk: Because of the above ex., $\mathcal{B}([0,1])$ is the σ -alg. generated by all open balls in $[0,1]$.

(from MK1)

4 Here is an exercise, to understand what kinds of sets are there in a σ -alg.

Exercise 13. Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space.

(1) $\mathcal{F}$ is closed under finite and countable unions, intersections, differences, symmetric differences. Also $\Omega \in \mathcal{F}$.

(2) If $A_n \in \mathcal{F}$, then

$$\limsup A_n := \{\omega : \omega \text{ belongs to infinitely many } A_n\},$$

$$\liminf A_n := \{\omega : \omega \text{ belongs to all but finitely many } A_n\}$$

are also in $\mathcal{F}$. In particular, if A_n increases or decreases to A , then $A \in \mathcal{F}$.

(3) $\mathbf{P}(\emptyset) = 0$, $\mathbf{P}(\Omega) = 1$. For any $A, B \in \mathcal{F}$ we have $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B) - \mathbf{P}(A \cap B)$. If $A_n \in \mathcal{F}$, then $\mathbf{P}(\cup A_n) \leq \sum \mathbf{P}(A_n)$.

(4) If $A_n \in \mathcal{F}$ and A_n increases (decreases) to A , the $\mathbf{P}(A_n)$ increases (decreases) to $\mathbf{P}(A)$.

5 One more ex: $\mathcal{B}([0,1]) = \sigma(\{[a,b] \cap [0,1]\}_{a < b})$.

6 Some other nice collections of sets. $\Omega \neq \emptyset$ set.

i) π -system S is a collection of subsets ^{of Ω} s.t.
 $A, B \in S \Rightarrow A \cap B \in S$.

ii) λ -system A collection of subsets $\Lambda \subseteq 2^\Omega$ is called a λ system if.

- $\Omega \in \Lambda$

- $A, B \in \Lambda$ where $A \subseteq B \Rightarrow B \setminus A \in \Lambda$

- $A_n \uparrow A$ and each $A_n \in \Lambda \Rightarrow A \in \Lambda$.

iii) Algebra $\mathcal{F} \subseteq 2^\Omega$ is an algebra if

- $\emptyset, \Omega \in \mathcal{F}$

- $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$

- $A, B \in \mathcal{F} \Rightarrow A \cup B \in \mathcal{F}$.

|| • For a σ -algebra, we have closure under countable union.

• But here, only finite union.

[7] Just like in the case of σ -algebras, one can talk about the λ -system or algebra generated by a collection $S \subseteq 2^\Omega$. These will be denoted by $\lambda(S)$ and $\mathcal{A}(S)$.

[8] Ex.

- Check that $S = \{(a, b] \cap [0, 1] : a < b\}$ is a π -system.

- What is the algebra $\mathcal{A}(S)$ generated by S ?

[9] Sierpinski - Dynkin π - λ thm.

(i) $\mathcal{F}$ is a σ -algebra $\Leftrightarrow \mathcal{F}$ is both a π -syst. & a λ -system.

(ii) If S is a π -system, then $\lambda(S) = \sigma(S)$.

Proof.

(i) $(\Rightarrow)$ is clear.

$(\Leftarrow)$ Let's assume $\mathcal{F}$ is both a π -syst. & λ -syst. and then prove that it is a σ -algebra.

- Since $\Omega \in \mathcal{F}$, if $A \in \mathcal{F} \Rightarrow \Omega \setminus A = A^c \in \mathcal{F}$.
- Let $\{A_n\}_{n=1}^{\infty} \subseteq \mathcal{F}$.

Take

$$B_n := A_1 \cup A_2 \cup \dots \cup A_n.$$

We claim that each $B_n^c \in \mathcal{F}$ (and hence $B_n \in \mathcal{F}$).

$$B_n^c = A_1^c \cap A_2^c \cap \dots \cap A_n^c$$

$\hookrightarrow$ each of these complements are in $\mathcal{F}$ as it is a λ -system.

- Intersections again in $\mathcal{F}$ as it is a π -system.

Now since $B_n \uparrow \left(\bigcup_k A_k\right)$ and each $B_n \in \mathcal{F}$ (which is a λ -system), we can conclude that $\bigcup_k A_k \in \mathcal{F}$.

(ii) It is clear that $\lambda(S) \subseteq \sigma(S)$. By (i), it follows that in order to prove $\lambda(S) \supseteq \sigma(S)$, it suffices to show that $\lambda(S)$ is a π -syst.

Want to show that

$$A \cap B \in \lambda(S), \quad \forall A, B \in \lambda(S).$$

$$S \cap \lambda(S) \subseteq \lambda(S)$$

Step 1: Fix $A \in S$, and define $\mathcal{C}_A$ by:

$$\mathcal{C}_A := \{ B \in \lambda(S) : A \cap B \in \lambda(S) \}.$$

We will show now that $\mathcal{C}_A = \lambda(S)$.

Notice first that $S \subseteq \mathcal{C}_A$. ($\because S$ is a π -system).

We now show that $\mathcal{C}_A$ is a λ -system:

- Clearly $\Omega \in \mathcal{C}_A$
- Suppose $B_1, B_2 \in \mathcal{C}_A$ s.t. $B_1 \subseteq B_2 \stackrel{?}{\Rightarrow} B_2 \setminus B_1 \in \mathcal{C}_A$.

$$A \cap (B_2 \setminus B_1) = \underbrace{(A \cap B_2)}_{\in \lambda(S)} \setminus \underbrace{(A \cap B_1)}_{\in \lambda(S)}.$$

So $A \cap B_1 \subseteq A \cap B_2$. & both in $\lambda(S)$

hence $A \cap B_2 \setminus A \cap B_1 \in \lambda(S)$

This proves that $B_2 \setminus B_1 \in \mathcal{G}_A$.

Step 2: $\lambda(S) \cap \lambda(S) \subseteq \lambda(S)$.

Let us fix $A \in \lambda(S)$ and def.

$$\mathcal{G}_A := \{B \in \lambda(S) : A \cap B \in \lambda(S)\}.$$

By step 1, $\mathcal{G}_A$ is a λ -system containing S .

Hence $\mathcal{G}_A = \lambda(S)$.

This proves that $\lambda(S) \cap \lambda(S) \subseteq \lambda(S)$ and hence $\lambda(S)$ is a π -system, which completes the proof of (ii).

Lemma Let P and Q be probability measures on $\sigma(S)$ where S is a π -system satisfying

$$P(A) = Q(A), \quad \forall A \in S.$$

Then $P = Q$.

Proof. Let $\mathcal{C} = \{B \in \sigma(S) : P(B) = Q(B)\}$.

By assumption, $S \subseteq \mathcal{C}$.

If we show that $\mathcal{C}$ is a λ -system, then

$$\lambda(S) \subseteq \mathcal{C} \subseteq \sigma(S)$$

||
 $\sigma(S)$ by previous thm. & we are done!

- Clearly $\Omega \in \mathcal{C}$.
- Suppose $B_1, B_2 \in \mathcal{C}$ s.t. $B_1 \subset B_2 \stackrel{?}{\Rightarrow} B_2 \setminus B_1 \in \mathcal{C}$

$$\begin{aligned} P(B_2 \setminus B_1) &= P(B_2) - P(B_1) = Q(B_2) - Q(B_1) \\ &= Q(B_2 \setminus B_1). \end{aligned}$$

So yes, $B_2 \setminus B_1 \in \mathcal{C}$.

- Assume $\exists$ an $\uparrow$ seq $B_n \uparrow B$ and each $B_n \in \mathcal{C}$.

From ex., we saw that

$$P(B) = \lim_{n \rightarrow \infty} P(B_n) = \lim_{n \rightarrow \infty} Q(B_n) = Q(B)$$

Thus $\mathcal{C}$ is actually a λ -system containing S .

$$\therefore \mathcal{C} = \lambda(S) = \sigma(S).$$
